

## OPERATING SYSTEM INFORMATION, WINDOWS, LINUX, MACOS

**1. On your Windows laptop or PC, open 'System Information' and take a screenshot showing your OS version, processor, and RAM details.**

**Ans.**

| Item                             | Value                                                                         |
|----------------------------------|-------------------------------------------------------------------------------|
| OS Name                          | Microsoft Windows 11 Home Single Language                                     |
| Version                          | 10.0.26200 Build 26200                                                        |
| Other OS Description             | Not Available                                                                 |
| OS Manufacturer                  | Microsoft Corporation                                                         |
| System Name                      | LAPTOP-LPFTBM4A                                                               |
| System Manufacturer              | ASUSTeK COMPUTER INC.                                                         |
| System Model                     | Vivobook_ASUSLaptop X1605ZA_X1605ZAC                                          |
| System Type                      | x64-based PC                                                                  |
| System SKU                       |                                                                               |
| Processor                        | 12th Gen Intel(R) Core(TM) i7-12700H, 2300 Mhz, 14 Core(s), 20 Logical Pro... |
| BIOS Version/Date                | American Megatrends International, LLC. X1605ZA.315, 1/25/2024                |
| SMBIOS Version                   | 3.5                                                                           |
| Embedded Controller Version      | 255.255                                                                       |
| BIOS Mode                        | UEFI                                                                          |
| BaseBoard Manufacturer           | ASUSTeK COMPUTER INC.                                                         |
| BaseBoard Product                | X1605ZA                                                                       |
| BaseBoard Version                | 1.0                                                                           |
| Platform Role                    | Mobile                                                                        |
| Secure Boot State                | On                                                                            |
| PCR7 Configuration               | Elevation Required to View                                                    |
| Windows Directory                | C:\WINDOWS                                                                    |
| System Directory                 | C:\WINDOWS\system32                                                           |
| Boot Device                      | \Device\HarddiskVolume1                                                       |
| Locale                           | United States                                                                 |
| Hardware Abstraction Layer       | Version = "10.0.26100.1"                                                      |
| User Name                        | LAPTOP-LPFTBM4A\mitpr                                                         |
| Time Zone                        | India Standard Time                                                           |
| Installed Physical Memory (RA... | 16.0 GB                                                                       |
| Total Physical Memory            | 15.7 GB                                                                       |
| Available Physical Memory        | 6.82 GB                                                                       |
| Total Virtual Memory             | 16.7 GB                                                                       |
| Available Virtual Memory         | 8.13 GB                                                                       |
| Page File Space                  | 1.00 GB                                                                       |
| Page File                        | C:\pagefile.sys                                                               |
| Kernel DMA Protection            | On                                                                            |
| Virtualization-based security    | Running                                                                       |

**2.Download any official Linux distribution ISO (like Ubuntu or Fedora), and use Rufus or BalenaEtcher to create a bootable USB drive for installation.**

**Ans.**

**Steps:**

- step.1 Download the Ubuntu Linux ISO file from the official website.
- step.2 Connect an 8 GB or larger USB drive to the computer.
- step.3 Open Rufus.
- step.4 Select the connected USB drive.
- step.5 Click Select and choose the downloaded ISO file.
- step.6 Keep the default settings.
- step.7 Click Start.
- step.8 Click OK to confirm that the USB data will be erased.
- step.9 Wait until Rufus finishes copying the files.
- step.10 When READY appears, the bootable USB is created.
- step.11 Safely remove the USB drive.
- step.12 The USB can now be used to install Ubuntu Linux.

**Result:**

Bootable Linux USB was successfully created using Rufus.

**3. On a macOS device (or using a macOS virtual machine), open 'System Preferences' and change the desktop wallpaper, then adjust the system language to Hindi or Gujarati.**

**Ans.**

**Steps:**

- step.1 Open the macOS device or macOS virtual machine.
- step.2 Open System Preferences from the Apple menu.
- step.3 Click on Desktop & Screen Saver.
- step.4 Select any wallpaper from the available options.
- step.5 The desktop wallpaper will be changed.
- step.6 Now go back to System Preferences.
- step.7 Click on Language & Region.
- step.8 Click the + (Add) button under Preferred Languages.
- step.9 Select Hindi or Gujarati from the language list.
- step.10 Add the selected language.
- step.11 Select the new language as the preferred language if required.
- step.12 Restart or log out of the Mac if prompted.
- step.13 Check that the selected language has been applied.

**Result:**

The macOS desktop wallpaper was changed and the system language was set to Hindi/Gujarati.

**4.Install a fresh copy of Ubuntu Linux (real or virtual machine) and create a new user account named 'gamer'. After installation, enable automatic login for this user and take a screenshot of the login screen showing the username.**

**Ans.**

**Steps:**

- step.1 Start the Ubuntu Linux installation using a USB drive or VirtualBox.
- step.2 Follow the installation instructions and complete the Ubuntu installation.
- step.3 After installation, open Settings.
- step.4 Go to Users.
- step.5 Click Unlock and enter the administrator password.
- step.6 Click Add User.
- step.7 Enter the username as gamer.
- step.8 Set a password for the new user and click Add.
- step.9 Log out from the current account.
- step.10 Log in once using the gamer account to check that it works.
- step.11 Open Settings → System → Users.
- step.12 Unlock the settings and select the gamer user.
- step.13 Enable Automatic Login for the gamer account.
- step.14 Restart the Ubuntu system.
- step.15 Check that Ubuntu automatically logs in with the gamer account.
- step.16 Take a screenshot showing the gamer username/login screen.

**Result:**

Ubuntu Linux was installed successfully, a new user named gamer was created, and automatic login was enabled for the user.

**5. Use ChatGPT to generate a step-by-step guide for installing Windows 10 on a laptop, including partitioning tips and post-installation settings for privacy and updates. Follow the guide and note any steps that were unclear or missing.**

**Ans.**

**Steps:**

- step.1 I used ChatGPT to get a step-by-step guide for installing Windows 10 on a laptop.
- step.2 I created a Windows 10 bootable USB drive.
- step.3 I restarted the laptop and booted it from the USB drive.
- step.4 I selected the language, time and keyboard settings.
- step.5 I clicked on Install Now.
- step.6 I selected the Windows 10 edition and accepted the license terms.
- step.7 I selected Custom: Install Windows only.
- step.8 In the partition section, I selected the required partition for Windows installation.
- step.9 I made sure not to delete any partition containing important personal data.
- step.10 I formatted the Windows installation partition and continued the installation.
- step.11 Windows copied the required files and restarted the laptop.
- step.12 After installation, I completed the basic Windows setup.
- step.13 I connected the laptop to the internet and checked for Windows Updates.
- step.14 I opened Settings → Privacy and reviewed the privacy options.
- step.15 I turned off unnecessary permissions such as location and app access where required.
- step.16 I checked Windows Update again and installed the available updates.
- step.17 I restarted the laptop after completing the updates.

**Unclear or Missing Steps:**

The exact partition size depends on the laptop and available storage.

Some Windows 10 setup screens may be different depending on the version.

Driver installation was not explained in detail, so I checked the laptop manufacturer's website if required.

**Result:**

Windows 10 installation steps were followed successfully, and the privacy and update settings were checked after installation.